

HACKATHON²⁰²⁶

Asset management

Applications of Agentic AI and Machine Learning in Portfolio Management

McGill-FIAM Asset Management
Hackathon Instructions

Presented by

FIAM



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Introductory note: Russ Goyenko (McGill University)

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We are excited to welcome you to the **third edition of the Asset Management Hackathon!** Building on two years of momentum, this competition continues to bridge the gap between **finance and data science** through a unique, cross-disciplinary format. Unlike traditional hackathons, our focus is on **Financial Economics** - exploring innovative applications of **technology, AI, and Big Data** to tackle one of the industry's most important challenges: **bottom-up portfolio construction**. This year we push the frontier one step further, into **Agentic AI**: systems that do not merely predict, but plan a research task, call tools, gather evidence, and act on it.

What's New This Year?

1. **U.S. Equity Focus:** The case narrows from last year's 28-country universe to **U.S. public equities**. **Teams receive the same monthly stock panel and a linked subset of company 8-K filings.**
2. **Market Neutral Mandate:** Every submission this year must be **equity market neutral** - long and short legs balanced so that the book carries little or no directional exposure to the market, inside a net exposure band of $\pm 50\%$. This is the strategy most popular among hedge funds and what institutional investors are looking for right now, and it changes what counts as success: you are no longer trying to beat the S&P 500, you are trying to beat **cash plus a premium**. The section below explains the mandate, the constraints and the benchmark in full.
3. **A More Recent Sample:** The panel is **monthly, from 01/2015 through 08/2026**, with the out-of-sample evaluation window running from **01/2021 to the most recent month available**. That test period contains the post-COVID rebound, the 2022 drawdown, the AI-driven concentration of 2023–2025, and everything since - a genuinely demanding stretch for any strategy.
4. **Text as Data (provided by [Quantillum](#)) — 8-K Filings:** Replacing last year's quarterly and annual reports, we provide **cleaned 8-K current reports** from U.S. public companies. **Most specified reportable events are subject to a four-business-day deadline, with exceptions described in the SEC's Form 8-K instructions. These company disclosures support event-driven signal discovery and provide evidence that an agent can read and cite.**
5. **Agentic AI:** For the first time, we invite you to build **agents** - LLM-driven systems that plan a research task, call tools, read filings, write and run their own code, and hand a defensible recommendation back to a portfolio construction step. A dedicated section below sets out the use cases we would like to see, and the failure modes we expect you to guard against.
6. **Expanded Industry Presence:** With even greater participation from **leading financial institutions**, students will gain exposure to recruiters seeking precisely the kind of technically skilled, finance-trained talent this event develops. Last year, almost all of the finalists were hired by our industry partners.

The Format

As with last year, the hackathon unfolds in **two stages**:

- **Stage One (Training Challenge):** Teams will develop portfolio trading strategies using the provided datasets, applying **machine learning and big data techniques**. To support you, we have included materials covering best practices in ML for stock selection. While you may incorporate external data, we strongly recommend using the provided datasets as they form part of the evaluation criteria.
- **Stage Two (Final Hack):** Based on Stage One submissions, the **top 10 teams** will advance to the finals. In this stage, participants will receive **additional training, full 2 day mentorship from industry experts who are real time portfolio managers, expanded text coverage, alternative datasets of the kind hedge funds actually buy, and direct exposure to leaders applying AI in asset management daily**. See the note on Carbon Arc later in this document.

This design ensures that even teams not advancing to the finals gain **practical, hands-on experience** with the foundational applications of AI/ML in asset management. We encourage you to engage faculty as academic advisors and to explore innovative approaches that push beyond conventional methods.

For inspiration, we invite you to visit the very first edition's event and outcomes: [Hackathon 2024 Archive](#). Consider it your benchmark to beat.

First Challenge Description and Supplementary Training Materials

Written Russ Goyenko (McGill) and Chengyu Zhang (Shanghai Jiao Tong University)

Introduction

The rapid growth of financial data—spanning stock characteristics, fundamentals, and textual sources—has transformed how investors build portfolios. With the rise of big data and advanced machine learning (ML), new opportunities have emerged to improve investment decisions and performance. The idea is simple: better data and smarter models can lead to better portfolios. What has changed most in the past year is not the models themselves, but who is able to use them — and how much of the research process they are now trusted to run.

Traditional financial models struggle with the scale and complexity of modern datasets. ML, however, excels at handling large, high-dimensional data, identifying the most relevant predictors, and reducing noise. Beyond quantitative signals, financial text—filings, press releases, management commentary—has become a powerful but underutilized source of insights. Advances in Large Language Models (LLMs) now make it possible to extract meaningful patterns from such unstructured text and link them to stock performance.

Until very recently, doing this at scale required a quantitative research desk. That barrier is falling fast. In September 2026 [The Wall Street Journal](#) described what it called the AI shift turning everyday investors into “mini quant funds”: retail investors describing a strategy in plain English, having an AI agent write and backtest the code, and routing the resulting orders straight into a brokerage account. The head of Moomoo U.S. told the paper that individual investors are becoming “mini hedge funds,” and expects agents to account for a meaningful share of retail order flow by the end of 2026. Robinhood, Webull and Public have all shipped features that connect an external model - ChatGPT, Claude, Grok - to a live account, some with a human in the loop and some without.

The same article is equally clear about the other side of the trade. Agent-generated strategies tend to crowd into the same concentrated, high-valuation names; they frequently fail to beat a passive benchmark once fees and taxes are counted; and millions of agents reading the same disclosures through the same handful of frontier models is a recipe for exactly the kind of correlated unwind the industry last lived through in the August 2007 quant meltdown. Fluency is not accuracy: a model will write a confident, well-argued investment thesis whether or not the underlying claim survives a backtest.

That tension - an enormous expansion of research capacity paired with genuine fragility - is the intellectual centre of this year’s competition. You are not being asked to build a chatbot that talks about stocks. You are being asked to build a **research process** that an agent can execute, that you can audit, and that survives strictly out of sample.

The Challenge

Your task is to design and test an investment strategy that leverages ML, LLMs, agentic workflows, or a combination of all three to:

1. Identify which stock-level factors are most predictive of future returns.
2. Construct a market-neutral U.S. equity portfolio, with the long and short legs balanced so that the book carries little or no net exposure to the market.
3. Evaluate performance through rigorous backtesting.
4. Demonstrate predictive power using strictly out-of-sample tests.
5. Document the workflow — including any agent's plan, tools, and prompts — so that a third party could reproduce it.

Your Tools

You'll work in Python, using a provided toolkit with:

- A monthly U.S. stock panel, 01/2015–08/2026: `chars_final_with_names.parquet` contains 529,082 stock-month observations.
- The 147 characteristics in `factor_char_list.csv`, plus 51 identifiers and auxiliary columns: 198 columns in total.
- A consolidated text file, `8k_20150101_20260831_identified.parquet`, containing 373,139 linked filing observations. Read `readme.md` for the schema, identifiers, text cleaning, and date conventions. Provider filing timestamps are preserved.
- Python teaching templates for model training and portfolio analysis. Adapt their CSV inputs, return-variable names, sample dates, and performance calculations to this release as explained below.

Your objective is to design a **market-neutral U.S. equity portfolio strategy** holding between 100 and 500 positions across both legs, within a gross exposure limit of 200% and net exposure band of –50% to +50%. You'll rebalance monthly, as our data come from monthly or quarterly frequencies. You can choose to rebalance less often but passive strategy is not sustainable for market neutral portfolios. Success will be judged against a **cash-plus-premium benchmark** — the 3-month U.S. Treasury bill rate plus 4% a year - and on **risk-adjusted metrics**, above all the **information ratio**, alongside alpha, drawdowns, turnover, and how close your realised beta actually sits to zero.

The main numerical dataset is `chars_final_with_names.parquet`. For each retained stock-month, it contains the 147 characteristics selected by the variable column in `factor_char_list.csv`, together with identifiers, dates, returns, prices, market capitalization, industry classifications, and label provenance. Use the CSV list to select the baseline predictors; the other 51 columns support merging, evaluation, and implementation and are not an additional characteristic list. The selected signals span fundamentals, past returns, liquidity, trading costs, and risk. Appendix A describes the characteristics and their research references; `readme.md` documents the supplied

files and joining conventions. Characteristic values and missingness are preserved from the source panel; preprocessing must be implemented and documented by each team.

Here is an example on how to prompt ChatGPT about signals:

You: What Bello and Lin 2011 tell about inventory growth for future stock return predictability

ChatGPT:



Bello and Lin (2011) explore the relationship between inventory growth and future stock returns. Their research suggests that inventory growth can be a predictor of future stock returns, particularly in the context of firms' operational efficiency and market expectations.

Key findings from Bello and Lin (2011) include:

1. **Negative Relationship:** They find a generally negative relationship between inventory growth and future stock returns. Firms with higher inventory growth tend to experience lower future stock returns. This is because rising inventories might indicate declining sales, inefficient operations, or potential overproduction, which are often viewed negatively by the market.
2. **Economic Interpretation:** The negative relationship is consistent with the idea that inventory growth signals declining demand or inefficiencies in production and inventory management, which the market tends to penalize. Conversely, firms that manage their inventories effectively or have inventory reductions often signal improving demand or operational efficiency, leading to higher future returns.
3. **Cross-Sectional Analysis:** The study's analysis is cross-sectional, meaning it examines how inventory growth impacts stock returns across different firms. The results suggest that inventory growth can be a useful factor in predicting cross-sectional differences in future stock returns.

Overall, Bello and Lin (2011) argue that inventory growth is a valuable indicator that investors can use to gauge future performance, particularly in assessing a firm's operational health and efficiency.

For a row labelled by eom (end-of-month) in month t, the characteristics belong to month t and ret_exc_lead1m is the stock's excess return in month t+1. Use this forward return as the target without shifting it a second time. The columns ret and ret_exc describe month t, not the next month. Do not include ret_exc_lead1m in the predictors. To use the legacy templates, map it to stock_exret and construct the template's date, year, and month from the target month t+1; preserve the original eom and source date separately.

What Do You Want to Predict?

The primary objective of this case is to **predict next month's stock returns**. If you can generate accurate forecasts, you would want to **go long** (buy) the stocks with the highest expected returns (e.g., top 100 names) and **short** the stocks with the lowest expected returns (e.g., bottom 100 names). Trades are executed at the beginning of the month, and results are evaluated at month-end. Profits are measured by portfolio **alpha** or simply the realized monthly returns of this long-short strategy. The Python script `portfolio_analysis_hackathon.py` in your toolkit provides a starter implementation; adapt its input schema, date alignment, and return accounting to the supplied Parquet files.

The Mandate: Equity Market Neutral

This year every submission must be **equity market neutral**. This is the strategy the hedge fund industry has converged on, and the reason is not hard to see: after a decade in which a passive S&P 500 position was very hard to beat, allocators stopped paying for market exposure they can buy for three basis points, and started paying only for the part of a manager's return that has nothing to do with the market.

A market-neutral portfolio holds longs and shorts in roughly equal measure, so that the two legs cancel out the market's direction. If the S&P 500 falls 10% in a month, a properly constructed neutral book should be close to flat: the shorts make what the longs lose. What survives is whatever your stock selection added - the longs outperforming the shorts. That residual is **alpha**, and for a market-neutral fund it is the entire product.

Neutrality has more than one definition, and you should be explicit about which one you enforce:

- **Dollar neutrality.** The dollar value of the long book equals the dollar value of the short book at each rebalancing date. Simple to implement and simple for us to verify.
- **Beta neutrality.** The beta-weighted exposure of the longs equals that of the shorts, so the portfolio's beta to the S&P 500 is approximately zero. Harder, and more honest - a dollar-neutral book that is long high-beta names and short low-beta names is not neutral at all, it is a levered long position wearing a disguise.
- **Sector or factor neutrality (optional refinement).** Neutralising within sectors, or against size, value and momentum, strips out further unintended exposures. This is what most professional market-neutral desks actually do, and it is where a lot of the craft lives.

In practice we ask you to run the book inside a **net exposure band of -50% to +50%**, with dollar neutrality as the natural centre of that band; the full set of constraints is in the Trading Criteria section below. The band exists so that you can express a modest directional view when your models genuinely have one - not so that you can run a long book with a hedge bolted on. We will estimate your realised beta ourselves. A strategy that sits at the edge of the band month after month, or whose beta to the S&P 500 is materially different from zero over the out-of-sample period, is a directional strategy however it was described in the deck, and will be evaluated as one.

Your Benchmark: Cash Plus a Premium

A market-neutral strategy cannot sensibly be benchmarked against the S&P 500, because it is not trying to track it. The right benchmark is the return you could have earned risk-free, plus the premium an allocator expects in exchange for hiring an active manager. For this competition:

$$\text{Benchmark} = \text{3-month U.S. Treasury bill rate} + 4\% \text{ per annum}$$

The cash leg matters more than students usually expect. A market-neutral book is nominally zero-cost — the short proceeds finance the longs — so the capital sitting as collateral earns the bill rate. With the 3-month bill around 3.7% in August 2026, the hurdle is roughly **7.7% annualised**; back in 2021, when bills yielded almost nothing, the same rule gave a hurdle close to 4%. Your benchmark is therefore a **time series, not a constant**, and you should compute it month by month across the out-of-sample period.

You can download the monthly series — TB3MS, 3-Month Treasury Bill Secondary Market Rate — directly from the Federal Reserve Bank of St. Louis: <https://fred.stlouisfed.org/series/TB3MS>

Mind the units. FRED reports TB3MS as an annualized percentage. Under the competition's monthly-rate approximation, the benchmark in decimal return units is $\text{TB3MS} / 100 / 12 + 0.04 / 12$. The supplied stock returns also use decimal units. The RF used to construct `ret_exc` and `ret_exc_lead1m` is not necessarily the same series as TB3MS; do not subtract the risk-free rate twice or assume those series cancel. Obtain market and cash-rate observations covering the full evaluation period.

We still ask you to show the S&P 500 alongside your strategy, for two reasons: it gives the reader context, and the regression of your excess returns on the market's excess returns is how we verify that your beta is close to zero. But the number that decides whether the strategy worked is your return against cash plus 4%, expressed as an information ratio. Define the active return in month t as your portfolio return less the benchmark,

$$\text{active}_t = R_{\text{portfolio, total}, t} - (\text{TB3MS}_t / 100 / 12 + 0.04 / 12)$$

and then

$$\text{Information Ratio} = \sqrt{12} \times \text{mean}(\text{active}_t) / \text{stdev}(\text{active}_t)$$

A weighted long–short stock–return spread and a total return on portfolio capital are different quantities. State your signed position weights, capital denominator, and cash/collateral convention. If portfolio total return equals the same benchmark cash return plus a portfolio excess return, active return simplifies to that excess return minus $0.04 / 12$. Otherwise retain the difference between the relevant cash-rate series. Apply the information-ratio formula to active returns, rather than to a regression intercept divided by residual volatility.

We will also read the Sharpe ratio and **maximum** drawdown. Compute the Sharpe ratio from portfolio returns in excess of the stated risk-free return; it should not give credit for earning that risk-free return alone. The information ratio against cash plus 4% additionally asks whether the strategy clears the competition's performance hurdle.

⚠️ **Two practical points.** First, shorting is not free. Borrow costs are real, hard-to-borrow names can be recalled

at the worst possible moment, and a short book concentrated in small, illiquid, expensive-to-borrow stocks may be untradeable. We are not asking you to model borrow costs explicitly, but we will read your short book with this in mind. Second, a market-neutral book earns smaller absolute returns than a long-only book, and that is the point — do not chase a headline number by quietly letting net exposure drift long. We will see it in the beta.

Universe and Sample Period

This year the universe is **U.S. public** equities. The supplied monthly characteristics panel runs from **01/2015** to **08/2026** and contains 6,561 distinct PERMNOs. The linked filing sample contains 3,687 distinct PERMNOs, all present in that panel. Text is not available for every stock or every month. Keep stock-months without a filing unless your strategy explicitly restricts its universe, and document that restriction. Out-of-sample portfolio evaluation runs from **01/2021** through 08/2026. The hedge fund industry evaluates success through **portfolio** returns, so your predictive model must translate into a trading strategy with measurable performance.

Alternative Targets: Predicting Fundamentals

Predicting stock returns directly is notoriously difficult. An alternative is to forecast **company fundamentals** that drive future returns. For example, firms that deliver **earnings or revenues above expectations** often experience stock price appreciation that extends over the following months.

Historically, analysts have been central in interpreting fundamentals, but even large institutions cannot cover every stock in the market. This is where **technology and ML models** can add significant value.

Selected accounting ratios in `chars_final_with_names.parquet` can also support fundamental forecasting; examples include `ebit_sale`, `niq_at`, and `niq_be`. Choose a future fundamental outcome, explain why it matters for returns, and document its construction and availability.

Predictions must be **forward-looking** and use only information available at the forecast date. For a quarterly fundamental target, explicitly identify the future reporting period and release date. A shift of three monthly rows in a carried-forward accounting panel does not automatically identify the next fiscal-quarter report.

Using External Data (Optional)

If your university has access to **WRDS** ([link](#)), you may supplement the provided datasets with additional sources. For instance, van Binsbergen, Han, and Lopez-Lira (2023) use 67 financial ratios (e.g., book-to-market, dividend yield) from WRDS to predict earnings per share (EPS).

If you merge external data, include:

- Data-cleaning code.
 - A clear description of the source.
 - Use a security identifier and a date: preferably permno with the relevant month, or the pair gvkey and iid with the relevant month. CIK is an issuer key and CUSIP can change; verify their effective periods rather than joining on an undated company name or ticker.
-

Text Data: 8-K Filings, or the Company News Feed from [Quantillum](#)

A key change this year is the switch from periodic filings to event filings. We provide a cleaned, linked subset of U.S. 8-K current reports for 01/2015–08/2026. SEC Form 8-K identifies reportable events and generally requires filing or furnishing within four business days, subject to item-specific exceptions ([see PDF file](#)). Detected item codes help identify the subject of a report:

- **Item 2.02 - Results of operations and financial condition, often associated with earnings announcements.** The release may be an exhibit rather than part of the supplied main text.
- **Item 1.01 / 1.02 - Material agreements:** contracts signed or terminated, credit facilities, partnerships, supply deals.
- **Item 2.01 / 2.05 / 2.06 - Transactions and write-downs:** completed acquisitions and disposals, restructuring charges, impairments.
- **Item 5.02 - Officer and director changes:** CEO, CFO and board turnover, including the abrupt kind.
- **Item 4.01 / 4.02 - Auditor changes and non-reliance:** a change of accountant, or an admission that previously issued financials can no longer be relied upon — two of the strongest distress signals in the entire corpus.
- **Item 7.01 / 8.01 - Regulation FD and other events:** guidance updates, buybacks, litigation, product announcements, and anything else management chooses to disclose.

These filings provide company disclosures behind many news reports. A single filing may cover several items, and its main text may refer to exhibits that are not included in this package. An agent can extract and compare the events described, while citing the document_id and supporting text. Treat the extracted items and dates as annotations to verify, not as guarantees of a single event or an exact publication time.

- The filing data are supplied in one Parquet file, not separate yearly files. Read selected columns or filter by filing_date to work with smaller subsets.
- Each retained filing has permno, gvkey, iid, cusip, cik, ticker, and company_name. Join to

chars_final_with_names.parquet using permno and a dated month key, cross-checking gvkey and iid. A separate cik_gvkey_linktable.csv is not required. Some verified security links have no characteristics observation in the exact filing month; retain or exclude those cases deliberately. See readme.md for the many-to-one join example and identifier-quality flags.

In the finals, top teams will gain access to additional text data.

Filing timing must respect the portfolio's information cutoff. A disclosure dated the 3rd cannot be used to form a portfolio on the 1st. In this release, filing_date is the provider date, filing_timestamp_utc is a date/midnight placeholder, and sec_accepted_at_utc is null. content_report_date describes the reported event and is not a release date. State a conservative availability convention; obtain independent acceptance-time evidence before making intraday or same-day execution claims. For the current monthly frequencies, you can accumulate the news and extract their sentiment within the current month t to make any forecasts for the month $t+1$.

How to Use LLMs with Text

The principle is simple: **use text as a predictor**. You can machine-encode the filings and feed them into ML models, either to forecast **returns** or **fundamentals**. The better your chosen model captures context, tone and materiality in these disclosures, the stronger your predictive power may be.

For example:

- Encode the body of each 8-K with a pretrained LLM (e.g., BERT, FinBERT), or have a general-purpose model summarize it into a structured record.
- Use the embeddings - or the extracted fields - as features in an ML model (e.g., XGBoost, LightGBM).
- Predict next-month returns, next-quarter EPS, or the sign and size of the post-event drift.

This is a simplified workflow, but it highlights the opportunity: combining **quantitative signals** with **textual insights** for a richer, more powerful model. For reference, see [Can AI Read the Minds of Corporate Executives?](#) (available on SSRN).

⚠ Keep it manageable: text modeling can become complex quickly. Start with simple setups before scaling.

Agentic AI: This Year's Frontier

An LLM used as an encoder is a feature extractor. An agent is something else: a model given a goal, a set of tools, and the freedom to decide in what order to use them. It plans, calls a tool, reads the result, revises the plan, and repeats until it can answer. In a research context the tools are the ones you already have - a dataframe of characteristics, a filing archive, a Python interpreter, a backtester.

The distinction that matters for this competition is this: a model that scores text produces a number. An agent produces a process. And a process can be inspected, replayed, criticised and improved — which is exactly what a research desk does with a junior analyst.

Use cases we would like to see

- **The analyst agent.** Give an agent a ticker, that month's 8-K filings for the company, and its recent characteristic history, and ask it to return a structured verdict: what happened, whether it was expected, and a signed conviction score. Run it across the cross-section and treat the output as a factor. This is what a junior analyst does; the open question is whether it survives out of sample.
- **The event-triage agent.** Not every 8-K matters. An agent that reads the item code and the body and distinguishes a routine dividend declaration from an auditor resignation is performing a classification that no fixed rule handles well, and it hands you a far cleaner event set to trade on.
- **The signal-discovery agent.** Point an agent at the 147 characteristics and let it propose, code and backtest candidate combinations. Judge it not on the best backtest it found, but on how its top-ranked idea holds up in a period it never saw. Be warned: an agent allowed to search freely will find spurious patterns faster than any human ever could. Multiple-testing discipline is your responsibility, not the model's.
- **The devil's-advocate agent.** Run a second agent whose only job is to attack the first one's thesis - find the disconfirming filing, the crowded position, the factor exposure that explains the alpha away. Ensembles of deliberately disagreeing agents are among the more promising ideas in this space.
- **The portfolio-construction agent.** Turn forecasts into positions subject to your constraints (100–500 names, 200% gross, net exposure inside $\pm 50\%$, a turnover budget, sector limits), and have the agent write down why each trade was made. Explainability is a deliverable here.
- **The reproducibility agent.** An agent whose job is to re-run your own pipeline and hunt for look-ahead bias, survivorship and leakage before you submit. We will be checking; you should check first.

⚠ **Two warnings, both taken seriously by professionals.** First, **agents are fluent whether or not they are right.** A well-written investment thesis is not evidence; the backtest is. Every agent-generated claim in your deck must be traceable back to data we gave you. Second, **agents crowd.** If millions of retail agents read the same filings through the same handful of frontier models, they converge on the same names, and those positions unwind together. If your strategy's edge is "ask a frontier model what it likes," assume several hundred other teams - and a fast-growing share of the market - have the same edge. The interesting work is in **what you feed the agent, how you constrain it, and how you verify it.**

You are not required to use agents. A well-executed classical ML strategy will beat a poorly conceived agentic one, and we will say so. But if you do build one, tell us what it did, show us the prompts and the tool definitions, and be honest about where it failed. A candid account of an agent that did not work is worth more to us than a polished account of one that supposedly did.

Bottom line: Whether you predict **returns directly**, forecast **fundamentals as leading indicators**, or integrate **text with quant data**, your challenge is to design a model that translates into a **profitable and explainable investment strategy**.

Choice of ML algorithms & Training:

New machine learning (ML) algorithms are being created almost every day. There are anywhere from a dozen to over a hundred algorithms available, depending on how they are improved. The choice of which algorithm to use is up to you. This is where your creativity comes in: based on your investment idea, which technology best supports your idea with data?

In finance, researchers often prefer supervised or semi-supervised machine learning, where we guide the machine on what to learn. For simplicity, we'll explain the most common approaches using basic linear ML methods. The discussions and code we'll provide are from Goyenko and Zhang (2022), following the foundational work by Gu et al. (2020).

As an example we implement the following machine learning methods: LASSO of Tibshirani (1996), Elastic Net (EN) of Zou and Hastie (2005), Ridge of Hoerl and Kennard (1970).

We analyze the predictive power of machine learning algorithms for stock returns. We therefore define a return on an asset in the most general form as:

$$r_{i,t+1} = E(r_{i,t+1}) + \epsilon_{i,t+1} \quad (1)$$

where:

$$E(r_{i,t+1}) = g^*(z_{i,t}) \quad (2)$$

Stocks are indexed as $i = 1, \dots, N_t$ and months as $t = 1, \dots, T$. Function $g^*(\cdot)$ represents the machine learning algorithm. It maintains the same form over time and across different assets, and leverages information from the entire panel. $z_{i,t}$ is what you feed the machine learning algorithm. In mathematical language, $z_{i,t}$ is a P dimensional vector of predictors (147 stock characteristics every month t). Equation (2) says that your machine learning algorithm attempts to predict the return of each stock for the next month, using information from the current month.

Tune hyperparameters using a validation sample, estimate model coefficients from training data, and reserve testing data for out-of-sample evaluation. `penalized_linear_hackathon.py` illustrates the linear-model and hyperparameter-search workflow, but it is a legacy teaching template. Replace `sample_data.csv` with a Parquet input or a deliberately constructed model table, map `stock_exret` as described above, and update its old 2000–2024 and eight-year training settings to the 2015-based schedule below. Fit learned preprocessing on admissible training information; do not allow target availability to define the prediction universe.

Performance evaluation:

Following Gu et al (2020), we first evaluate the statistical performance of predictability by calculating the out-of-sample, OOS , R^2 as

$$R_{OOS}^2 = 1 - \frac{\sum_{(i,t) \in \mathcal{T}_3} (r_{i,t+1} - \hat{r}_{i,t+1})^2}{\sum_{(i,t) \in \mathcal{T}_3} r_{i,t+1}^2}$$

$\mathcal{T}_3$ denotes the test observations in the equation $r_{i,t+1}$ is the realized return, and $\hat{r}_{i,t+1}$ is the forecast for the same stock and holding month. Note that we don't subtract the historical mean in the denominator like you would normally see for out-of-sample R^2 . The implication here is that the benchmark we compare our predictions against is not the historical mean of the stock returns, but rather zero, which means there is no predictability for stock returns at all. This is because relying on the historical mean to construct a portfolio actually delivers worse performance than simply randomly selecting stocks in the market. It also means that, if you achieve a positive out-of-sample R^2 , no matter how small it is, you are capturing some predictability in the market. In fact, the out-of-sample R^2 of stock returns are not impressive, usually ranging between 1% and 2%, even with more complex models such as neural networks. The OOS R^2 calculation appears near the end of `penalized_linear_hackathon.py`; apply it to the correctly aligned target return and forecast pairs.

In your own analysis and in the submission deck, please do provide $OOS R^2$ of the methodology you are using.

Training Procedures:

Use an expanding training window and a rolling two-year validation window, assigning all splits by the target return month. For the first forecast year, use available target months in 2015–2018 for training, 01/2019–12/2020 for validation, and 01/2021–12/2021 for testing. Because the characteristics begin in January 2015, the first target month supplied by `ret_exc_lead1m` is February 2015; do not invent December 2014 predictors. For the next forecast year, expand training through 12/2019, roll validation to 01/2020–12/2021, and test 01/2022–12/2022. Continue through 08/2026, refitting annually and forecasting monthly. A January 2021 return forecast uses December 2020 characteristics. Keep labels whose realization falls in the test period out of training and validation, even if their characteristic month precedes the test boundary. Adapt the expanding-window block in `penalized_linear_hackathon.py`; its existing dates and line-number references are not the 2026 competition specification.

Trading Strategy Portfolio Evaluations:

While the statistical performance provides some preliminary ideas about the accuracy of the predictions, investors care more about the economical benefits that the model provides. That is, whether we can use the predictions to construct a portfolio that generates superior returns. A simple long-short strategy can be built by first sorting the stocks based on the predicted returns. At the beginning of every month (time t), we rank the stocks from low to high by their predicted returns. We then equally divide the stocks into ten buckets/portfolios, where the first portfolio contains stocks we predict to perform the worst, and the 10th portfolio contains the stocks we predict to perform the best. We would buy the best/top portfolio and short-sell the worst/first portfolio, assuming our holding of each stock is of the same dollar amount (equal weights), and check the performance of this strategy at the end of that holding month. Note that this decile spread is, by construction, a dollar-neutral

portfolio whose beta still needs to be checked, and that it has zero cost: the long positions are fully financed by the money we get from the short positions. We repeat this procedure for each month in the out-of-sample period. An alternative strategy would be to simply buy 100 stocks for which we have the highest predicted returns, and short sell 100 stocks for which we have the lowest predicted returns. The decile-ranking block in `portfolio_analysis_hackathon.py` illustrates this step. Adapt it to the target return month, enforce the position and exposure limits, and state the capital normalization.

Evaluate portfolio returns, Sharpe ratio, alpha, beta, information ratio against cash plus 4%, maximum loss, drawdown, and turnover. `portfolio_analysis_hackathon.py` is a starter reference that requires adaptation: its variable `stock_exret` must refer to the target holding month; its printed Information Ratio is an alpha-to-residual-volatility ratio, not this competition's benchmark information ratio; and its drawdown and turnover examples must be reconciled with the definitions used in your submission. The template does not enforce the 100–500-position or exposure constraints for you.

Estimation of portfolio Alpha and Beta:

We define the alpha as the intercept from a simple linear regression where you regress your portfolio excess over risk free rate returns on the excess over risk free rate returns of S&P500. That is

$$R_{p,t} - r_{f,t} = \alpha + \beta(R_{SP500,t} - r_{f,t}) + \epsilon_t$$

Regress portfolio excess returns on S&P 500 excess returns for the same holding months. The supplied target is `ret_exc_lead1m`, called `stock_exret` only after adapting the template of supplied python code examples; `ret` is a raw stock return and `ret_exc` is a same-month excess return. Construct portfolio total and excess returns consistently with your signed weights and cash/collateral convention before regression. Ask any AI assistant to download monthly S&P500 returns from FRED St. Louis, that's where you get risk free rates as well.

The intercept is Alpha or risk adjusted return, and the slope coefficient is the beta. These concepts are normally covered in the Introductory finance classes so we can skip further elaborations. When you want to annualize Alpha (as an output from the regression it is monthly), you can multiply it by 12.

For a market-neutral strategy this regression does double duty. The slope is your evidence of neutrality - over the full out-of-sample period it should sit close to zero, and we will read a beta far from zero as a failure of the mandate rather than a stylistic choice. The intercept, meanwhile, should account for very nearly all of your return, since there is no market exposure left to explain it. Report both, and report the standard error on beta as well as on alpha.

Compute resources

A quick GPU access for free is available at [LightningAI](#). You can use their platform to open a free account which comes with ~35h of free GPU per account per month. There is also Google Cloud trial and AWS free tier for students that allow you an access to free GPUs if you run deep learning models.

Looking Ahead to the Finals: Alternative Data from Carbon Arc

A heads-up for teams thinking about Stage Two. The 147 characteristics in your toolkit are, almost without exception, derived from prices and from accounting statements - the same raw material every quantitative manager has had since the 1990s. The competitive frontier moved elsewhere some time ago. What separates a modern fundamental-quant desk from a textbook one is **alternative data**: information about what a company is actually doing right now, observed from outside its own reporting.

The examples are easy to picture. Credit-card and receipt panels that show a retailer's revenue building week by week, six weeks before the earnings release. Web traffic and app downloads. Job postings that reveal where a firm is quietly expanding. Healthcare claims. Point-of-sale and supply-chain records. Shipping manifests, foot traffic, weather. The appeal is obvious: these series are high frequency, they are **observed rather than reported**, and they arrive before the fundamentals they predict.

In the finals we expect to give the top teams access to alternative datasets from **Carbon Arc** (<https://www.carbonarc.co/>), a New York firm that operates an Insights Exchange — a marketplace that standardises alternative datasets from many providers into a common, model-ready structure, connects them to real-world entities such as companies and locations, and delivers them through an API priced by consumption rather than by annual contract. Its co-founder and CEO built the research operations at Point72 and Glenview Capital, and the categories the platform aggregates - transaction and receipt panels, web and app activity, healthcare claims, point-of-sale and supply-chain records, labour and weather series — are the same ones the large funds are buying today. Historically this kind of data was available only to firms able to sign six- and seven-figure contracts and staff a team to clean it. Putting it in students' hands is precisely the point.

Extra prizes. Carbon Arc representatives will be at the finals, and they currently consider (no guarantees for now) awarding additional prizes of their own for the most creative use of their data - in return forecasting, in signal construction, or in portfolio management. These are judged separately from the main competition: a team can win a Carbon Arc prize without placing in the overall standings, and the criterion is the originality of the application rather than the size of the backtested Sharpe ratio. If you have ever wanted to test an idea that needs data nobody publishes, this is the moment for it.

How to prepare now. You cannot use this data in Stage One and you should not design your submission around it. But two habits will pay off if you reach the finals. First, build your pipeline so that a new block of features can be merged in on a firm-month key without rewriting everything downstream - most teams lose their first day in the finals to a join. Second, think in advance about which of your signals is weakest, and where knowing what a company is really doing would help you most. Teams that arrive with a specific question to ask of the data get far more out of it in short 2 days than teams that arrive hoping to browse.

Discussion

It's important to remember that we're providing you with a basic toolkit, and that within the market-neutral mandate there is still a great deal of freedom. How you size positions (equal weight, conviction weight, volatility scaling), which definition of neutrality you enforce (dollar, beta, sector, factor), how often you rebalance, and whether you scale gross exposure up or down as conditions change are all yours to decide. If you do vary gross or net exposure over time, make sure to explain what drives the decision to "switch" and how you predict it in advance — but gross exposure stays at or below 200% and net exposure stays inside the -50% to +50% band throughout, and you should be able to say why the book was wherever it was.

Whatever strategy you choose, it's crucial that you avoid using forward-looking information. In other words, you can't use actual events from month $t+1$ to make decisions in month t . That's why, along with your final submission decks, we're asking you to submit your Python codes. We'll be checking to make sure there's no misuse of forward-looking information in your strategy.

This applies with equal force to agents, in a way that is easy to miss. A frontier model trained on data through 2026 already "knows" what happened to a stock in 2021; asking it for a view on that period is not forecasting, it is recall. Constrain the agent to the documents and characteristics available at time t , and state explicitly in your deck how you enforced that constraint. Teams that cannot explain how they ruled out model-side look-ahead will be treated as having used it.

It is also important to recognize that we give you a start-up toolkit for ML methodology and how we apply it to finance data, and the codes with further details. Purely replicating our codes for the final submission decks will not be sufficient. You can get started with them, see what kind of performance you obtain with pure linear algorithms, and then build on it with your improvements.

Trading Criteria

- **Number of positions:** at least 100 and at most 500 names, counting long and short positions together.
- **Gross exposure:** up to 200% of capital - equivalently, leverage of at most 2x.
- **Net exposure:** between -50% and +50% of capital, at every rebalancing date.
- **Reporting:** report your average and maximum gross exposure, your average and extreme net exposure, and your realised beta to the S&P 500 over the out-of-sample period.

Beyond the exposure limits above, you have no trading restrictions except rebalancing your portfolio. You can rebalance once per month, once per quarter or half a year. In the toolkit and examples provided we rebalance every month and report monthly portfolio turnover. The objective here is an active portfolio management strategy. At the very least you have to rebalance some (not all) of your holdings once every half-year.

In plain English. Suppose you are managing \$100.

- **Gross exposure** is the total size of everything you hold, longs and shorts added together as absolute values. A 200% limit means those positions cannot sum to more than \$200 - for instance \$100 long and \$100 short.
- **Leverage of 2x** is the same constraint seen from the other side: you may put at most twice your capital to work. In a long-short book it is the short proceeds that finance it.
- **Net exposure** is longs minus shorts - how much of the market's direction you are actually carrying. A band of -50% to +50% means the two legs must stay within \$50 of each other. \$125 long against \$75 short is +50% net, the most bullish tilt allowed; \$75 long against \$125 short is -50% net, the most bearish. \$100 against \$100 is exactly neutral, and that is the centre of the band and the natural home for this mandate.
- **100 to 500 names** counts both legs together. Sixty longs and sixty shorts is 120 positions and is fine; forty and forty is not. The floor is there to stop you concentrating into a handful of bets; the ceiling is there because a book of a thousand names is a closet index fund with extra trading costs.

A worked example within the rules: \$110 long across 150 names and \$90 short across 130 names. Gross exposure is \$200 (200%), net exposure is +\$20 (+20%), and you hold 280 positions. All three constraints hold.

Measurement Frequency: Monthly Selection, Daily Risk

The data we provide are **monthly**, because most of the signals in the toolkit are monthly: characteristics are constructed from accounting and price data at a monthly frequency, and your predicted variable is next month's return. Selection is therefore a monthly decision, and the backtest we ask for is a monthly one.

But once you have chosen the month's positions, nothing stops you from **measuring** the portfolio daily - holding the month's weights fixed and marking the book every trading day. We strongly encourage you to do this, because monthly returns hide a great deal of what actually happened.

Consider a month in which the book is down 6% by the middle of the month and claws its way back to finish flat. On monthly data that month contributes a return of zero: no volatility, no drawdown, nothing to see. On daily data it is a 6% drawdown, and the volatility and Value-at-Risk computed from the daily path are far higher than anything the monthly series suggests. Both descriptions are arithmetically correct. Only one of them tells you what it would have felt like to hold the position — and only one of them tells you whether a prime broker would have made a margin call along the way, because margin is calculated on the daily path, not the monthly one.

This is why risk teams and allocators measure daily. A strategy whose monthly numbers look serene but whose daily path is violent is a different product from one that is genuinely calm, and the difference shows up in exactly the statistics an investment committee asks about: drawdown, volatility, and VaR. If you report daily-based figures alongside the monthly ones, say so clearly and label which is which.

Where to get daily prices: two routes, both realistic for a student team.

- **WRDS**, if your university has a subscription — CRSP daily stock files give you split- and dividend-adjusted daily returns for the full U.S. universe, and are the cleanest source available to you.
- **The Alpha Vantage API** as an alternative, at <https://www.alphavantage.co/> — a free API key covers the daily time-series endpoints (TIME_SERIES_DAILY, and TIME_SERIES_DAILY_ADJUSTED on the paid tier). Check the current limits on request rates and history depth before you build around it, and cache what you pull rather than re-querying.

To be clear about the weighting: **monthly results and analysis will be considered equally**. A team that submits a careful monthly analysis is not penalised, and a daily analysis will not rescue a weak strategy. Daily measurement is a way to show the committee that you understand what your monthly numbers are concealing - which is a differentiator, not a requirement.

Reports

Designing an investment strategy is the holy grail of the quantitative asset management industry. Most initial attempts either fail or prove to be non-tradable in real life due to market frictions. These frictions will prevent you from achieving high performance. For example, if you have to rebalance 100% of your portfolios every single month, most of your positive performance (also known as Alpha) will be erased by trading costs.

However, it does not matter at this stage, as it is a constant struggle for all quantitative investment professionals. What matters is to implement an idea and discuss the idea, and what you learned from it and what potential improvements can be done.

Therefore, we want you to give it your best effort and present your most promising idea, and its execution with the data we provide. Your creativity in setting up the training and the methodology, and how you use the data will be important criteria for evaluation.

Guidelines for the Deck

Please use PowerPoint slide format. Here is what to include:

Page 1

Executive Summary

Summarize your strategy, the ML algorithm(s) and/or agentic workflow chosen, and portfolio performance against the market-neutral benchmark (3-month T-bill + 4%), with the S&P 500 shown for context. Use market and benchmark data covering the complete 01/2021–08/2026 evaluation window.

Page 2

Describe your investment strategy: how you construct the long and short legs, and which definition of neutrality you enforce. What predictive signals you use to form the strategy. Present your **top 10 long and top 10 short holdings** on average over the OOS testing period, 01/2021 to 08/2026, identifying every position by **ticker and full company name**. Plot cumulative performance returns of your trading strategy vs the benchmark (3-month T-bill + 4%) and vs the S&P500 for OOS testing period, 01/2021 to 08/2026

Identify every holding, chart label, and table row by ticker and company name, for example “NVDA, NVIDIA Corporation”. Use permno and dated gvkey–iid relationships for merging. Period-supported ticker/name labels are included, but some characteristics observations remain unlabeled. For any such holding in the deck, verify its historical display labels from an authoritative source and record that source; do not substitute a later company name without verification.

Page 3

Data and Methodology: describe the data and methodology. Try to justify why you chose a specific ML (LLM) approach. How do you structure your training? If there is any new model architecture or training approach you introduce – please describe it here. If you chose to supplement our data with extra data (it is optional but we would be happy to see it) – please describe the data, whether they turned out to be valuable signals or not. Also here is the place to present OOS R2 statistics for the overall sample for ML algorithms you used.

If any part of your pipeline is agentic, describe the architecture here: what the agent could see, which tools it could call, how many steps it was allowed, which model you used, and how you prevented look-ahead on the model side. Put the prompts and tool definitions in the Appendix.

Page 4 to 7

Portfolio Performance statistics for OOS testing period, 01/2021 to 08/2026 for your portfolio vs the benchmark (3-month T-bill + 4%) and vs the S&P 500

Treat this as the performance pack you would hand to an investment management committee. Report every statistic below for the strategy and, where meaningful, for the benchmark and S&P 500 over 01/2021–08/2026. `portfolio_analysis_hackathon.py` provides selected teaching examples, not a complete implementation of these requirements. Implement the stated benchmark information ratio, drawdown, exposure, and turnover definitions explicitly and report any additional assumptions.

Return statistics

- Average monthly return and annualized return, both arithmetic and geometric (CAGR).
- Cumulative return over the full out-of-sample period.
- Calendar-year returns, year by year, for the strategy, the benchmark and the S&P 500.
- Best month and worst month, with dates.
- Hit rate: the share of months with a positive active return.
- Return of the long leg and the short leg reported separately, so the committee can see which side is actually working.

Risk-adjusted performance

- Information ratio, annualized — the headline number, computed as mean active return divided by the standard deviation of active return, times $\sqrt{12}$.
- Sharpe ratio, annualized, reported for completeness.
- Annualized alpha versus the S&P 500, with its t-statistic.
- Beta versus the S&P 500, with its standard error — your evidence of neutrality.
- Correlation of your monthly returns with the S&P 500.

Exposure, concentration and implementation

- Average number of holdings, split between the long and short legs.
- Average gross exposure and average net exposure, each with its range over the period.
- Average and maximum single-position weight, and the share of the book held in the top 10 names.
- Average monthly turnover with its range
- Characteristics of the short book: average market capitalisation, average dollar volume, and the share of short positions in small-cap or plausibly hard-to-borrow names.

Charts, at minimum

- Cumulative return of the strategy against the benchmark and the S&P 500.
- Underwater (drawdown) chart for the strategy, with the S&P 500 overlaid. Rolling 12-month active return and rolling 12-month information ratio.
- Rolling 12-month beta to the S&P 500 — the single clearest picture of whether you stayed neutral throughout, or only on average.
- Histogram of monthly returns with the benchmark's monthly hurdle marked.
- The 10 largest positive and 10 largest negative contributors to P&L over the period, by ticker and company name.

Page 4 to 7 might not hold all of this. Put the headline statistics table and three or four charts on the pages, and move the remaining tables and exhibits into the Appendix - that is precisely what the Appendix is for. Label every table and chart with the period it covers and state whether returns are gross or net of trading costs. What we are looking for is evidence that you can produce, and read, the pack an investment committee would actually be handed.

Page 8

The discussion of your strategy. Did it perform the way you trained it and did it meet your expectations? What are the main fundamental signals contributing to the performance of your portfolio? What are the most profitable positions (stocks) that drove the performance, and why. What are the macro-economic events that contributed to the performance. Potential improvements that you could make to this strategy. Identify every stock you name by ticker and full company name.

Following the 8 pages deck, *you can attach an Appendix not exceeding another 10 pages*. You are free to put in Appendix anything you think will help us to evaluate your work better, or any details that you could not include in the main deck. Please feel free to use any visuals that can help the presentation.

FINAL SUBMISSION PACKAGE

1. Your deck with Appendix in one PDF file. Please use power point slide format and then convert it to pdf file.
2. **IMPORTANT:** CSV file with your Monthly STOCK Holdings for Long and Short Positions for each month of testing period 01/2021–08/2026. The columns are Date (beginning of each month, ex. Sept 01 2026), PERMNO, TICKER, COMPANY NAME, WEIGHT (% of portfolio NAV invested in each stock for this month, Positive for Long and Negative for Short Positions). Another separate CSV file with the time series of your total portfolio monthly (or daily if you have) returns for the testing period.
3. Your Python code – please submit one file called MAIN.py which clearly describes your process. Please **DO NOT SUBMIT THE DATA** to run this code. The only data you need to submit is the monthly holdings in #2 above.
4. Licence – there is no licence as we are not creating a new technology. The idea is to find new ways, new ideas of applying existing technologies to financial data. Therefore, *your license (ex. Apache), Hackathon submission = public domain / free*. It is your idea backed up by back tests performance that will be evaluated by selection committee.
5. Prepare CVs for each team member – it is a part of submission, and the final team members' registration. Your CVs will be used for recruiting purposes by our industry participants, so please make sure they are up to date.

Our Evaluation Criteria

Selection committee comprise of industry experts and finance academics will be evaluating your project based on two main things: your investment idea (which financial factors you're focusing on and predicting) and the machine learning tools you choose to bring this idea to life.

You can get creative with your idea. We've suggested selecting stocks based on return predictions, but you could also predict other factors like earnings surprises, price-earnings ratios, or price-sales ratios that might drive future returns. The more original your idea, the more we'll appreciate your effort.

You can also innovate with the technology. We've provided basic examples using simple linear ML algorithms, which are easy to understand but may not capture the full complexity of financial data. You can explore alternatives — feed-forward neural networks, gradient boosting, transformer-based text models — or go further and build an agentic research pipeline of the kind described above. Trying out different technologies is part of your innovation and contribution. What we reward is not the sophistication of the tool but the quality of the reasoning around it: why this technology, for this signal, on this data.

Lastly, we'll compare your portfolio's performance to the market-neutral benchmark — the 3-month T-bill plus 4% a year - and we will check that the book really is neutral before we look at anything else. The S&P 500 is shown alongside for context, but beating it is not the objective this year and a high beta will not be read as a strength. We'll consider your portfolio's performance, but the main focus will be on your originality, choice of technology, and overall execution of the back-test.

Remember, it's not just about having the best return-what matters is your unique idea, the tools you use, and how well you put it all together.

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Appendix A.

A.1 Stocks-Specific Features

Table 1: Stock-specific Features

Feature	Acronym	Reference
Firm age	age	Jiang Lee and Zhang (2005)
Liquidity of book assets	aliq_at	Ortiz-Molina and Phillips (2014)
Liquidity of market assets	aliq_mat	Ortiz-Molina and Phillips (2014)
Amihud Measure	ami_126d	Amihud (2002)
Book leverage	at_be	Fama and French (1992)
Asset Growth	at_gr1	Cooper Gulen and Schill (2008)
Assets-to-market	at_me	Fama and French (1992)
Capital turnover	at_turnover	Haugen and Baker (1996)
Change in common equity	be_gr1a	Richardson et al. (2005)
Book-to-market equity	be_me	Rosenberg Reid and Lanstein (1985)
Market Beta	beta_60m	Fama and MacBeth (1973)
Dimson beta	beta_dimson_21d	Dimson (1979)
Frazzini-Pedersen market beta	betabab_1260d	Frazzini and Pedersen (2014)
Downside beta	betadown_252d	Ang Chen and Xing (2006)
Book-to-market enterprise value	bev_mev	Penman Richardson and Tuna (2007)
The high-low bid-ask spread	bidaskhl_21d	Corwin and Schultz (2012)
Abnormal corporate investment	capex_abn	Titman Wei and Xie (2004)
CAPEX growth (1 year)	capx_gr1	Xie (2001)
CAPEX growth (2 years)	capx_gr2	Anderson and Garcia-Feijoo (2006)
CAPEX growth (3 years)	capx_gr3	Anderson and Garcia-Feijoo (2006)
Cash-to-assets	cash_at	Palazzo (2012)
Net stock issues	chcsho_12m	Pontiff and Woodgate (2008)
Change in current operating assets	coa_gr1a	Richardson et al. (2005)
Change in current operating liabilities	col_gr1a	Richardson et al. (2005)
Cash-based operating profits-to-book assets	cop_at	
Cash-based operating profits-to-lagged book assets	cop_atl1	Ball et al. (2016)
Market correlation	corr_1260d	Assness, Frazzini, Gormsen, Pedersen (2020)
Coskewness	coskew_21d	Harvey and Siddique (2000)
Change in current operating working capital	cowc_gr1a	Richardson et al. (2005)
Net debt issuance	dbnetis_at	Bradshaw Richardson and Sloan (2006)
Growth in book debt (3 years)	debt_gr3	Lyandres Sun and Zhang (2008)
Debt-to-market	debt_me	Bhandari (1988)
Change gross margin minus change sales	dgp_dsale	Abarbanell and Bushee (1998)
Dividend yield	div12m_me	Litzenberger and Ramaswamy (1979)
Dollar trading volume	dolvol_126d	Brennan Chordia and Subrahmanyam (1998)
Coefficient of variation for dollar trading volume	dolvol_var_126d	Chordia Subrahmanyam and Anshuman (2001)
Change sales minus change Inventory	dsale_dinv	Abarbanell and Bushee (1998)
Change sales minus change receivables	dsale_drec	Abarbanell and Bushee (1998)
Change sales minus change SG&A	dsale_dsga	Abarbanell and Bushee (1998)
Earnings variability	earnings_variability	Francis et al. (2004)
Return on net operating assets	ebit_bev	Soliman (2008)
Profit margin	ebit_sale	Soliman (2008)
Ebitda-to-market enterprise value	ebitda_mev	Loughran and Wellman (2011)
Hiring rate	emp_gr1	Belo Lin and Bazdresch (2014)

Table 1 continued from previous page

Feature	Acronym	Reference
Equity duration	eq_dur	Dechow Sloan and Soliman (2004)
Net equity issuance	eqnetis_at	Bradshaw Richardson and Sloan (2006)
Equity net payout	eqnpo_12m	Daniel and Titman (2006)
Net payout yield	eqnpo_me	Boudoukh et al. (2007)
Payout yield	eqpo_me	Boudoukh et al. (2007)
Pitroski F-score	f_score	Pitroski (2000)
Free cash flow-to-price	fcf_me	Lakonishok Shleifer and Vishny (1994)
Change in financial liabilities	fml_gr1a	Richardson et al. (2005)
Gross profits-to-assets	gp_at	Novy-Marx (2013)
Gross profits-to-lagged assets	gp_atl1	
Intrinsic value-to-market	intrinsic_value	Frankel and Lee (1998)
Inventory growth	inv_gr1	Belo and Lin (2011)
Inventory change	inv_gr1a	Thomas and Zhang (2002)
Idiosyncratic skewness from the CAPM	iskew_capm_21d	
Idiosyncratic skewness from the Fama-French 3-factor model	iskew_ff3_21d	Bali Engle and Murray (2016)
Idiosyncratic skewness from the q-factor model	iskew_hxz4_21d	
Idiosyncratic volatility from the CAPM (21 days)	ivol_capm_21d	
Idiosyncratic volatility from the CAPM (252 days)	ivol_capm_252d	Ali Hwang and Trombley (2003)
Idiosyncratic volatility from the Fama-French 3-factor model	ivol_ff3_21d	Ang et al. (2006)
Idiosyncratic volatility from the q-factor model	ivol_hxz4_21d	
Kaplan-Zingales index	kz_index	Lamont Polk and Saa-Requejo (2001)
Change in long-term net operating assets	lnoa_gr1a	Fairfield Whisenant and Yohn (2003)
Change in long-term investments	lti_gr1a	Richardson et al. (2005)
Market Equity	market_equity	Banz (1981)
Mispricing factor: Management	mispricing_mgmt	Stambaugh and Yuan (2016)
Mispricing factor: Performance	mispricing_perf	Stambaugh and Yuan (2016)
Change in noncurrent operating assets	ncoa_gr1a	Richardson et al. (2005)
Change in noncurrent operating liabilities	ncol_gr1a	Richardson et al. (2005)
Net debt-to-price	netdebt_me	Penman Richardson and Tuna (2007)
Net total issuance	netis_at	Bradshaw Richardson and Sloan (2006)
Change in net financial assets	nfna_gr1a	Richardson et al. (2005)
Earnings persistence	ni_ar1	Francis et al. (2004)
Return on equity	ni_be	Haugen and Baker (1996)
Number of consecutive quarters with earnings increases	ni_inc8q	Barth Elliott and Finn (1999)
Earnings volatility	ni_ivol	Francis et al. (2004)
Earnings-to-price	ni_me	Basu (1983)
Quarterly return on assets	niq_at	Balakrishnan Bartov and Faurel (2010)
Change in quarterly return on assets	niq_at_chg1	
Quarterly return on equity	niq_be	Hou Xue and Zhang (2015)
Change in quarterly return on equity	niq_be_chg1	
Standardized earnings surprise	niq_su	Foster Olsen and Shevlin (1984)
Change in net noncurrent operating assets	nncoa_gr1a	Richardson et al. (2005)
Net operating assets	noa_at	Hirshleifer et al. (2004)
Change in net operating assets	noa_gr1a	Hirshleifer et al. (2004)
Ohlson O-score	o_score	Dichev (1998)
Operating accruals	oaccruals_at	Sloan (1996)
Percent operating accruals	oaccruals_ni	Hafzalla Lundholm and Van Winkle (2011)
Operating cash flow to assets	ocf_at	Bouchard, Krüger, Landier and Thesmar (2019)
Change in operating cash flow to assets	ocf_at_chg1	Bouchard, Krüger, Landier and Thesmar (2019)

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Feature	Acronym	Reference
Asset tangibility	tangibility	Hahn and Lee (2009)
Tax expense surprise	tax_gr1a	Thomas and Zhang (2011)
Share turnover	turnover_126d	Datar Naik and Radcliffe (1998)
Coefficient of variation for share turnover	turnover_var_126d	Chordia Subrahmanyam and Anshuman (2001)
Altman Z-score	z_score	Dichev (1998)
Number of zero trades with turnover as tiebreaker (6 months)	zero_trades_126d	Liu (2006)
Number of zero trades with turnover as tiebreaker (1 month)	zero_trades_21d	Liu (2006)
Number of zero trades with turnover as tiebreaker (12 months)	zero_trades_252d	Liu (2006)